

Zeyad Elshamy

AI & Robotics Engineer

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Regensburg, Germany (Open to relocation)

SUMMARY

AI & Robotics Engineer focused on autonomous systems and perception, building deep learning and computer vision pipelines in Python (PyTorch/TensorFlow) and integrating them into ROS2 workflows. Experienced in simulation-driven development and validation (Gazebo), real-time robustness, and delivering reproducible systems (Docker, Git). Published research on a humanoid personal assistant robot and motivated to build deployable, high-performance autonomy solutions

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Embedded C, JavaScript, HTML, CSS
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, CNNs
- **NLP:** Transformers, Embeddings, Tokenization, sentence-transformers (SBERT), ROUGE, BERTScore
- **Computer Vision:** OpenCV, YOLO, MediaPipe, Object Detection, Tracking
- **Autonomy/Robotics:** ROS1/ROS2, Gazebo, SLAM, Sensor Fusion, Path Planning (basic), Kinematics
- **Backend/APIs:** REST APIs, FastAPI, OpenAPI/Swagger
- **Data & Tools:** NumPy, Pandas, Plotly, Git, GitHub, GitHub Actions (CI/CD), Linux, Docker, VS Code, AWS, EasyEDA
- **Hardware/Protocols:** NVIDIA Jetson, Raspberry Pi, Arduino, LattePanda, I2C, UART, SPI
- **Productivity:** PowerPoint, Word, Excel, Outlook

PROJECTS

Advanced Text Paraphrasing Engine (NLP)

12/2025 – Present

Personal Project

Python, NLP, FastAPI, Docker, sentence-transformers (SBERT)

- Developing a secure paraphrasing service exposing a FastAPI REST API and containerized deployment with Docker.
- Implementing N-best paraphrase generation + ranking using SBERT semantic similarity, copy-rate penalty, and configurable decoding.
- Adding PII masking & restoration (emails, phone numbers, IPs, IDs) to support privacy-preserving text processing.
- Building an evaluation workflow (e.g., BERTScore/ROUGE + qualitative review) to validate meaning preservation and fluency.

Autonomous Lane Detection System

11/2025 – 12/2025

Personal Project

Python, OpenCV, ROS2, Gazebo

- Crafted a lane-perception “sense layer” that turns raw camera frames into stable lane-boundary estimates for autonomous driving scenarios.
- Combined classical vision and geometric reasoning (Canny, Hough Transform, curve fitting) to produce smooth, frame-consistent lane tracking.
- Integrated the pipeline into ROS2 + Gazebo to evaluate behavior in simulated roads and iterate quickly with modular nodes.

Eye State and Drowsiness Detection System

10/2025 – 11/2025

Personal Project

Python, TensorFlow, OpenCV, MediaPipe, pytsx3

- Built a real-time driver vigilance system that fuses CNN-based eye-state recognition with yawning detection (MAR) and head-pose cues.
- Designed a temporal drowsiness scoring strategy to filter noisy signals and trigger reliable fatigue warnings.
- Delivered an end-to-end safety experience with live visual overlays and text-to-speech audio alerts.

Humanoid Robot “Rafeqi”

09/2023 – 09/2024

Graduation Bachelor's Project, AASTMT

Python, ROS, OpenCV, TensorFlow

- Developed a humanoid personal assistant robot focused on human-friendly interaction and autonomous operation in complex environments.
- Integrated perception and intelligence (computer vision, NLP) with motion execution using kinematics for precise, repeatable behaviors.
- Published the design and implementation in “RAFEQI: Design and Implementation of a Cost-Effective Personal Assistant Robot” (RIMC, AASTMT Publishing Centre) – Published on 06 March 2025.

RESEARCH

Publication: RAFEQI: Design and Implementation of a Cost-Effective Personal Assistant Robot
Robotics: Integration, Manufacturing & Control (RIMC), AASTMT Publishing Centre
• Published on 06 March 2025.

03/2025
Published Paper

EDUCATION

Technische Hochschule Deggendorf (THD), Campus Cham 03/2025 – Present
M.Eng. AI for Smart Sensors & Actuators Cham, Germany
• **Focus:** Advanced Programming, Advanced Sensor Technology, AI and Machine Learning, System Design, Model-Based Function, Edge-Device Architecture.

Arab Academy for Science, Technology & Maritime Transport (AASTMT) 08/2020 – 09/2024
B.Sc. Artificial Intelligence (Robotics and Intelligent Systems) – Final Grade: 3.0/4.0, Dean's List Alamein, Egypt
• **Relevant courses:** Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Image Processing, Advanced Robotics.

LEADERSHIP & ACTIVITIES

AAST Robotics Club (Egypt) – Hardware Committee Member 2021 – 2024
Student Club Egypt
• Contributed to robotics hardware prototyping and system assembly, collaborating with teammates on sensor/actuator integration and testing.

ICPC Final Competition (Egypt) – Organization Committee Volunteer 2023
Volunteer Egypt
• Supported on-site competition operations, including logistics coordination, participant assistance, and event flow management.

CERTIFICATIONS & TRAINING

- Introduction to Cloud 101 – AWS Training and Certificate (2024).
- Machine Learning Foundations – AWS Training and Certificate (2024).
- Artificial Intelligence and Applications – Zewail City of Science and Technology (2023).
- Summer School in Computer Science and AI – Autonomous University of Barcelona (UAB) (2022).
- RoboCup Junior Egypt – Rescue Line Category Participant (Egypt, 2022).

SOFT SKILLS

- Analytical thinking and structured problem solving
- Clear communication and technical presentation
- Team collaboration in cross-functional environments
- Rapid learning and adaptability to new tools/tech stacks
- Creative thinking and managing parallel priorities
- Hands-on hardware and electronics proficiency

LANGUAGES

- Arabic – Native (C2)
- English – Fluent (C1)
- German – A2 (currently improving)